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Planetary Rover

Heterogeneous Multi-Robot Exploration

SUBJECT: Artificial Intelligence for Robotics II (AI4RO2)

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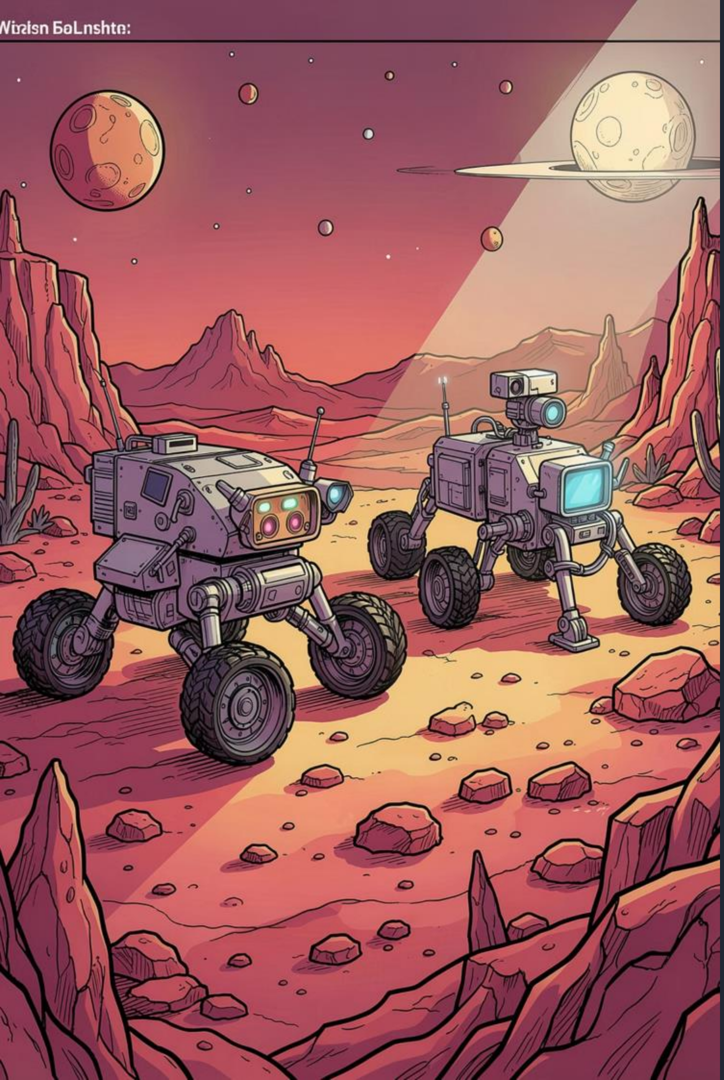


Domain & Planning Problem Overview

Phase 1: Classical STRIPS

Classical planning to validate logical coordination and task allocation between heterogeneous rovers.

Phase 2: PDDL+

Hybrid Dynamical System with continuous time, numeric fluents, and autonomous environmental dynamics.

Multi-agent cooperation emerges organically from spatial and terrain constraints via the **can-traverse** predicate — no explicit synchronization actions required.

Part 1: Types, Predicates & Actions

Core Predicates

Component Type	Element Name & Parameters	Description & Purpose
Type	rover, location, sample, terrain	Defines the hierarchical object classes used in the domain to ensure type safety.
Predicate	(at ?r ?l)	Tracks the current spatial location ?l of a specific rover ?r.
Predicate	(connected ?l1 ?l2)	Defines the graph topology (navigable edges) between two locations.
Predicate	(terrain-type ?l ?t)	Assigns a specific terrain type ?t (e.g., flat, rough) to a location node ?l.
Predicate	(can-traverse ?r ?t)	**Crucial for Heterogeneity:** Defines hardware capabilities. Maps which rover can drive on which terrain, avoiding duplicate action definitions.
Predicate	(sample-at ?s ?l)	Tracks the physical location of a sample when it is on the ground.
Predicate	(carrying ?r ?s)	Indicates that rover ?r is currently holding sample ?s in its gripper.
Predicate	(empty ?r)	Indicates the rover's gripper is free. Required to pick up a new sample.
Predicate	(delivered ?s)	The **Goal State**. Indicates the sample has been successfully secured at the base.
Predicate	(is-base ?l)	Flags a specific location node as the designated drop-off point for the mission.

Actions

Action Name	Preconditions (What must be true)	Effects (State changes)	Purpose & Discussion Context
Maps	- Rover is at ?from - ?from and ?to are connected - Target has terrain-type - Rover can-traverse target terrain	- (not (at ?r ?from)) - (at ?r ?to)	Moves the rover. Note: Uses capability-based constraints instead of hardcoding separate wheeled/legged actions.
collect-sample	- Rover is at sample location - Sample is sample-at location - Rover gripper is empty	- (not (sample-at)) ground - (not (empty)) gripper - (carrying) sample	Allows the rover to pick up a target item. Enforces payload capacity (1 item at a time).
drop-sample	- Rover is at location - Rover is carrying sample	- (not (carrying)) - (empty) gripper - (sample-at) ground	Key for Cooperation: Allows a rover to leave a payload at an intermediate zone for another rover to pick up (Relay Handover).
deliver-sample	- Rover is at location - Location is-base - Rover is carrying sample	- (not (carrying)) - (empty) gripper - (delivered) sample	Fulfills the mission objective. Strictly requires the rover to be at the designated base station.

Terrain types (flat, rough, mixed) gate mobility via **can-traverse**, avoiding duplicate action definitions.

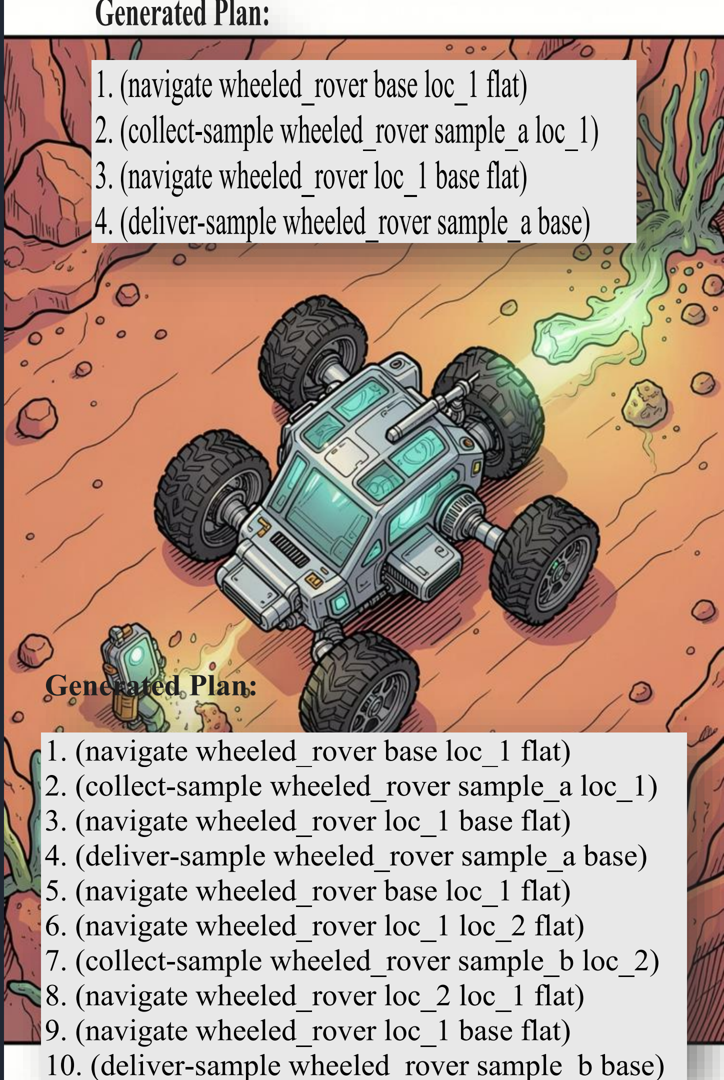
Problems 1 & 2: Single-Robot Scenarios

Problem 1 — Single Sample

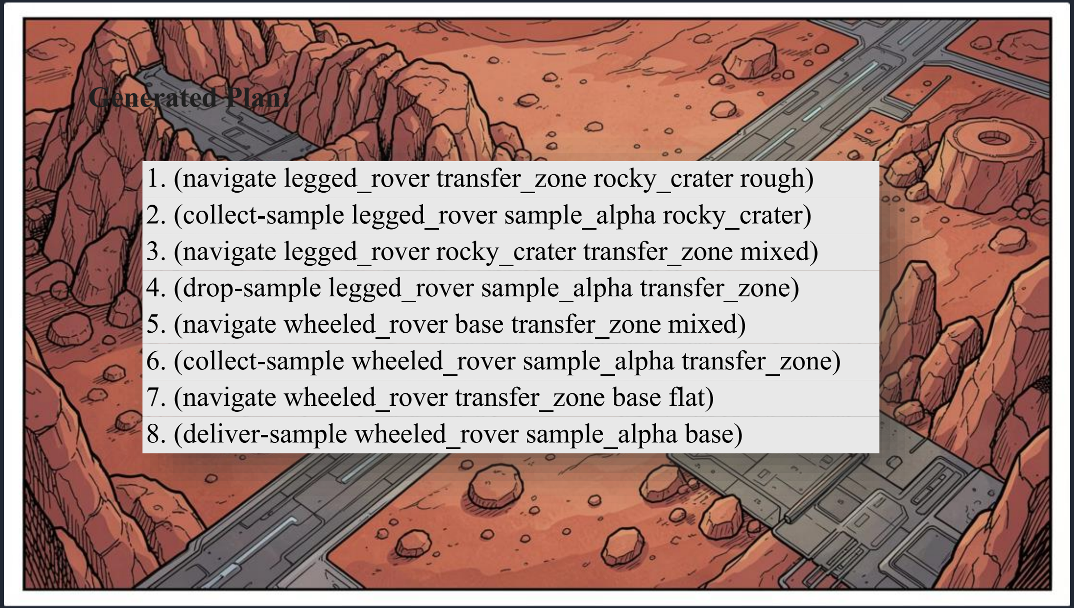
Wheeled rover retrieves one sample from flat terrain and delivers to base. ENHSP generates a minimal **4-step optimal plan**.

Problem 2 — Multiple Samples

Rover retrieves two samples across expanded terrain. Planner handles repetitive allocation, completing sample_a's full cycle before initiating sample_b retrieval — a **10-step sequential plan**.



Problem 3: Forced Cooperation



A sample in rough terrain is **inaccessible to the wheeled rover**; the legged rover cannot traverse flat terrain to base. The planner resolves this via a **relay handover** at the mixed-terrain **transfer_zone**.

01

Gatherer (Legged)

Navigates rough terrain, collects sample, drops at transfer_zone

02

Courier (Wheeled)

Collects at transfer_zone, navigates flat terrain to base

Problem 4: Advanced Coordination

A branched crater network with two samples in disconnected rough zones. The planner distributes workload based entirely on mechanical constraints.

Phase 1 Mining

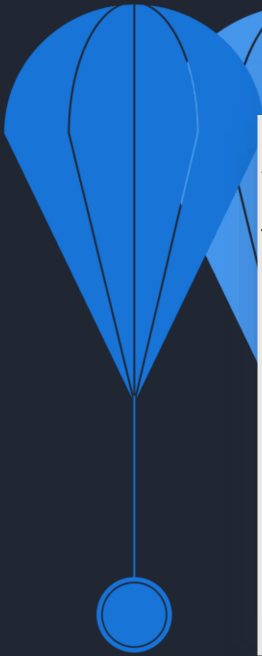
Legged rover mines north and south craters

Sample Deposit

Deposit collected samples at central hub

Phase 2 Delivery

Wheeled rover shuttles samples hub-to-base



Generated Plan:

1. (navigate legged_rover hub crater_south rough)
2. (collect-sample legged_rover sample_s crater_south)
3. (navigate legged_rover crater_south hub mixed)
4. (drop-sample legged_rover sample_s hub)
5. (navigate legged_rover hub crater_north rough)
6. (collect-sample legged_rover sample_n crater_north)
7. (navigate legged_rover crater_north hub mixed)
8. (drop-sample legged_rover sample_n hub)
9. (navigate legged_rover hub crater_south rough)
10. (navigate wheeled_rover base hub mixed)
11. (collect-sample wheeled_rover sample_n hub)
12. (navigate wheeled_rover hub base flat)
13. (deliver-sample wheeled_rover sample_n base)
14. (navigate wheeled_rover base hub mixed)
15. (collect-sample wheeled_rover sample_s hub)
16. (navigate wheeled_rover hub base flat)
17. (deliver-sample wheeled_rover sample_s base)

The legged rover acts as a dedicated **miner**, the wheeled rover as a **logistics courier** — cooperation emerges purely from terrain constraints.

Part 2: PDDL+ Hybrid Dynamical System

Element Type	Name & Parameters
Predicate	(path-terrain ?l1 ?l2 ?t)
Predicate	(is-navigating ?r) / (is-resting ?r)
Predicate	(is-broken ?r)
Function	(nav-progress ?r)
Function	(risk-level ?r)
Function	(max-risk) / (speed) / (risk-rate)

Action Name	Key Preconditions	Key Effects
start-navigate	- (path-terrain) matches (can-traverse) - not (is-broken) - not (is-navigating)	- Flags (is-navigating) - Assigns (nav-progress) = (distance)
end-navigate	- (is-navigating) is True - (nav-progress) <= 0.0	- Clears (is-navigating) - Updates (at ?to)
start-rest	- Target is not (is-rough) - (risk-level) > 0.0 - not (is-broken)	- Flags (is-resting)
stop-rest	- (is-resting) is True	- Clears (is-resting)
Manipulation Actions	- not (is-navigating) - not (is-resting) - not (is-broken)	- Updates (carrying) / (empty)

Processes (Continuous)

- **Maps-process** — distance reduction at speed × #t
- **accumulate-risk** — hazard buildup in rough terrain
- **recover-risk** — risk depletion during rest

Events (Autonomous)

- **risk-failure** — fires when risk-level exceeds max-risk, sets is-broken and speed=0

Navigation Design

Durative actions avoided. Navigation split into instantaneous **start/end** actions modulated by continuous processes.

PDDL+ Problems 1 & 2: Continuous Dynamics

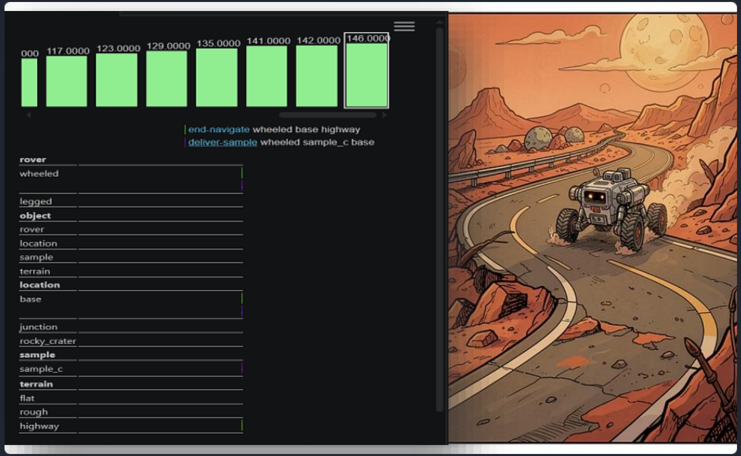
Problem 1 — Baseline

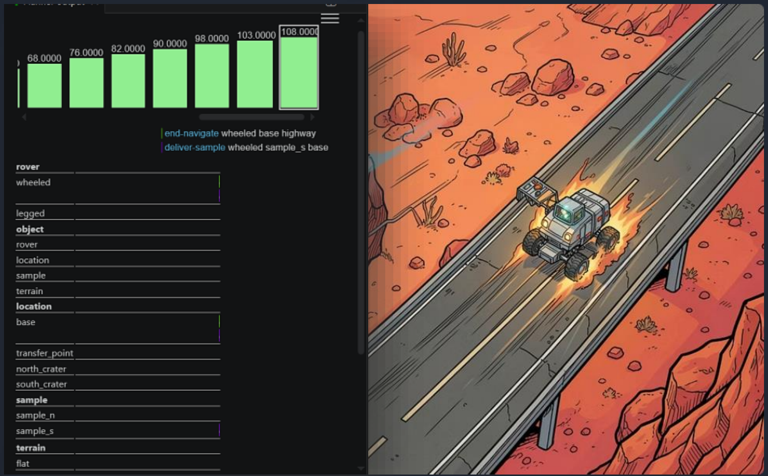
Single-agent scenario validating distance depletion and risk accumulation. ENHSP synthesized a **16-step plan, 10.0 time units**. Risk stayed below threshold — no resting required.

Problem 2 — Proactive Risk Mitigation

Planner forced to use **safe_island** for mandatory rest. Each rough leg generates 45 risk; a full round-trip would exceed max-risk (100). The planner autonomously inserted rest phases — a **107-step plan, 77.0 time units**.

PDDL+ Problems 3 & 4: Multi-Agent Hybrid Plans





Problem 3 — Time-Critical Cooperation

Legged rover gathers from rough terrain; wheeled rover exploits **highway** for fast logistics. **202 steps, 146.0 time units.**

Problem 4 — Scaled Optimization

Concurrent scheduling with **(:metric minimize total-time)**. Maximum parallelization — legged mines while wheeled delivers. **152 steps, 108.0 time units.**

Results, Limitations & Conclusion

Key Achievements

- STRIPS: Emergent cooperation via **can-traverse** capability encoding
- PDDL+: Continuous navigation, risk accumulation, and autonomous **risk-failure** event
- Planner dynamically schedules mandatory rest to prevent systemic failure

Limitations

- PDDL+ temporal state-space explodes with asynchronous parallel agents
- **TMP Gap:** Symbolic terrain abstraction ignores slope gradients, traction, and kinodynamic constraints
- Real-world deployment requires integration with geometric motion planners (e.g., RRT)

This work highlights the trade-off between spatial pruning through heterogeneity and temporal state-space explosion in hybrid planning.